

Lec 20 Intuition behind KL Divergence - Part B

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Transcript

[00:00] Hello everyone. Welcome to the lecture series on fundamentals of generative AI and the large language models. We are in the week three. In this particular recession, we will come to an end for KL divergence where we will see self-information, entropy, cross-entropy, and a detailed derivation of KL divergence. Before you watch this particular session, you have to finish the previous one. Then you will have a continuity and you'll be able to understand the concept well. So, first let us see what exactly is a self-information.

[00:32] Now, let us take one example. What is the example? We want to check what is tomorrow's weather. We want to predict tomorrow's weather. What are the two possible outcomes for tomorrow's weather? Either it can be sunny or it can be rainy. Now, you will ask the model to give a prediction. So, the model has a predicted that around closely around to 90% or around 87% there is a chance that it is sunny. Around 12% there is a chance that it is rainy.

[01:08] This is what model has predicted. So, let me give an example I mean repeat again. You have an Your task here is to predict tomorrow's weather. The possibilities are either it can be sunny or rainy. Now, you have trained the model and you have

asked the model to give a prediction. So, the model has a predicted that around 87% there is a chance that it becomes it is sunny. Around 12% it is rainy. This is what model has predicted. Once the model has a predicted, now you are seeing what is tomorrow's weather. The first possibility is tomorrow is sunny only.

[01:45] Tomorrow it turned out to be sunny. When it turned out to be sunny, are you surprised? Definitely no, right? Why you are not surprised? Because you were already sure that the model has a predicted 87% and it turned out to be sunny only. So, I'm not very much a surprised. So, here if tomorrow is a sunny, then you are not very much surprised. Why you are not very much surprised? Because you were already knowing it. Why you are knowing it?

[02:13] Because model was very confident that it is 87% that it is going to be sunny. So, here you are not very much surprised. Why you are not very much surprised? Because the model prediction is what has occurred. What is the other possibility? The other possibility is what if tomorrow it turns out to be rainy, right? If tomorrow is a rainy, then you are very much surprised. Why you are very much surprised? Because this is something which is very less likely to happen. Rain was very unlikely, but it turned out to be rain.

[02:48] So, from this example, what is that we are measuring? Is it not we are measuring how much you are surprised, right? So, self-information is all about it measures how surprised one particular outcome is. Okay? So, for an outcome X, what is X here? Is it not the sunny is one X, rainy is one X? For for an outcome X and with this probability P of X, self-information is defined as I of X equal to $\log 1 / P$ of X. This is the formula for self-information. What is self-information? It is all about measuring how surprising one particular outcome is. Now, when you say sunny is the outcome, is it not the sunny is one outcome, right? X and then it is also having a probability that is P of X. So, for a outcome X and with the probability

[03:51] P of x, self information is uh defined as i of x uh $\log 1 / p$ of x. Okay. Now, uh from the basic fundamental of log, uh anything is $1 / a$, can we write it as $\log 1 / a$, can we write it as $-\log a$? That is what has been written here. So, i of x is $-\log p$ of x. So, here why it is? Because $\log 1 / p$ of x is $-\log$. So, now, if what is the relation here? Can you look into the relation between these two? Here, you are very much less surprised. Surprising factor is denoted as i of x. Why you are less surprised?

[04:39] Because the probability the model's probability was very high. When model's probability was very high, the surprise factor was very less. When model probability is very less, sunny was very less, but it turned out to be sunny. Therefore, surprise is very large. So, therefore, you look into if p of x is a large, then i of x is a small, which means less surprise. When p of x is a small, then i of x is increasing, then you have more surprise.

[05:09] So, is it not it is inversely proportional? Because it is inversely proportional,

you are writing as $1 / p$ of x . I hope how the formula is obtained you have understood. What we have discussed here is a self information. Let us see one example uh of this self information. Before we look into the example, you can ask the question, why did the log come here? What is the role of log here? When suppose when two independent events happen, always log can two independent events can we take it as AB? Then, log will convert AB product into an summation. Log AB AB can be written as log A plus log B. So, two independent events log will convert multiplication into addition. So, because of this, we are using a logarithm. So, let us see

[06:07] now under log while log 2 when log 2 is used, then you can measure in terms of bits. Self-information is a measured in bits. Now, let us take an example. The probability that the model predicts that it is a sunny is 0.875 or 87.5% confidence or the probability is 0.875. If this is your P of X , then what is the I of X ? I of X formula is minus log 2 of P of X . So, substitute the value in this and you have got 0.193 bits. What is this 0.193? The unit what we are using here is a bits.

[06:53] Similar way, rainy probability is 0.125. If that is the case, what is its a surprise? Surprise is very large. Why surprise is very large? Because it is inversely proportional. So, this is what is one simple example and whatever is a number that we are getting, that number measures the surprise and it is a giving one you know particular outcome. Measuring the surprise of one particular outcome. If you look into the rain carries more.

[07:25] Now, from this what what information it is inferred? Is it not rain carries more self-information? Why rain carries more self-information? Because that is something which is less likely to happen. Therefore, self-information has increased. Right? So, self-information measures the surprise of one outcome only. This is all about the self-information. Second the term what we have is entropy. So, what is self-information? One particular outcome. I of X is minus log two of P of X . So, now this is all about self-information, but what we are measuring here is about the entropy.

[08:08] Now, aim here is to predict tomorrow's weather. That is the X . X is tomorrow's weather. So, what are the two possible outcomes? The two possible outcomes are sunny and rainy. These are possible outcomes. So, we have still not seen tomorrow. If we have not seen these are the two possible, sunny and rainy. Before tomorrow arrives, we do not know which outcome is going to occur. Which outcome will occur. There is some uncertainty, definitely, because the possibilities are these two. And there is some uncertainty here.

[08:44] Why uncertainty? It can be either sunny or it can be rainy. Now, this is what is the model of prediction. Point 0.875 for sunny and 0.125 is for the rainy. So, which means the model is saying 87% chance of sunny weather, 12% 12.5% of rainy weather. Together, it is a model's predicted probability distribution. Okay? Now, what we are discussing is all about the entropy. What is the entropy? Entropy considers

all possible outcomes and their probability also.

[09:28] And it is going to measure the overall uncertainty. Right? Now, if it is a still not a clear, let us look into this particular. What is the self information says? Self information describes one observed outcome only. But, we want one number. That one numerical number should tell what is the overall uncertainty of the entire probability distribution. We want one single number which will tell the uncertainty of the overall. So, then we require an entropy. So, this is a self information. I of X is a self information.

[10:10] How it is, we have already seen $-\log_2 p(x)$. But, before the actual outcome is known, there are several outcomes are possible, right? So, what entropy will do? Entropy combines the surprise of all of these outcomes into one average value. That is denoted as $h(p)$. So, it is entropy is given as $h(p)$, and it is given by the formula summation of $p(x) \log_2 p(x)$. Just substitute what is $I(x)$ formula? Is it not the $-\log_2 p(x)$?

[10:51] So, substitute that. In the place of $I(x)$, you have substituted $-\log_2 p(x)$. Take this a minus a term outside. Then you get this. Then, we have also seen if you don't want minus, can we write it as $1/p(x)$? So, if you don't want the minus a term, you can also write it as $1/p(x)$. So, this is what is the formula. But, let us understand with an example so that it will be clear to you. Now, what is the $h(p)$? $h(p)$ says that you have to combine all of the you know, outcomes, its a surprise, and its a probability. So, its a surprise is given as $1/p(x)$. So, is it not this is giving you the surprise and this is the probability. So, just substitute the values. Come It It combines all of it,

[11:45] which means the outcomes are two. So, there are two terms. One is for sunny, one is for the rainy. So, by combining this, we are getting one value 0.544. What is this value? Now, what is entropy? For entropy, we want to one number. And this number will tell what is the uncertainty here. Now, we got that one number. From this number, what are we understanding? Here entropy value is very low, 0.5. The entropy value is low.

[12:18] When the entropy value is low, see, 0.544 is $h(p)$. What is $h(p)$? $h(p)$ is an entropy. And that entropy value is low. When entropy value is uh relatively low, what is the indication? The indication is one outcome is much more likely than the other. Right? Here the model has a predicted 87 and other one is a 12. So, definitely one outcome is more likely to occur. That is what is the information we are getting from this entropy. From entropy, what is that we are understanding? Because entropy value is very low. What low?

[13:00] 0.544. When it is very low, we are able to understand this because one event is more likely to occur than the other event. That is what is the information we are getting. Suppose, instead of 87.5 and a 12.5, what if the model has a given 0.5 and

0.5? If you If model has a predicted 50% chances of uh sunny, 50% chances of rainy, yeah, what is uncertainty here? Is it not uh first of all, when you substitute the values in this formula, we get H of P value very high. But that high value means H of P value was one.

[13:44] Why one? There is a maximum uncertainty. Why maximum uncertainty? The result is mostly unpredictable. You cannot predict. It can be either sunny or it can be rainy. So, is it not from this H of P value being high, you're saying that it is maximum uncertainty. Look at other case. Here uh the model has a predicted zero probability for sunny, 100% for rainy. Then what should be your H of P value? H of P value is zero. Why it is zero? There is no uncertainty at all. Why there is no uncertainty? Uh then the weather is completely you can predict it. Because the model is saying 100% So, we can completely predict the weather.

[14:28] So, this is one single number we are getting. From this a single number, we are measuring what is the uncertainty. How did you get this a single number? You got this a single number by considering both [clears throat] outcomes, its surprise and its probability. So, its a surprise is I of X and its probability is P of X. And you have to summation here is a considering both the outcomes. Right? This is entropy. Entropy is a mainly measuring the um you know, overall uncertainty before actually the event occurs. So, it multiplies surprise of each outcome by its probability and adds the result. So, before observing the outcome, how uncertain is the whole situation is entropy. Entropy is a measuring of the overall uncertainty. I

[15:27] hope this is clear to you. Now, the third term we have a cross entropy. Cross entropy, before we actually see the definition, let us understand with an example, okay? Then from this example, we'll connect to an explanation of cross entropy. Now, let us assume now there are two outcomes. The two outcomes are one is sunny, second one is rainy. Now, the target suppose the actual outcome is sunny. The actual outcome is sunny. Now, there is a predicted probability. What is that predicted probability is a Q. The model has given the prediction. So, let us assume that we are denoting it as a Q.

[16:11] Okay? And what is that the model has a predicted? The model has a predicted it as 0.875 and 0.125. Sometime back we have seen together, if you look into to Yeah. So, together if you see, we call it as model's predicted probability distribution, right? So, together we are calling it as probability distribution only. So, here let us assume we have This is what is the model prediction. 0.875 and 0.125 together is model's probability distribution, predicted probability distribution.

[16:48] Then, what are two possible outcomes? Sunny and rainy. What is the true distribution? P. Is it not if it is actually sunny, sunny should get one and rainy is zero? So, this is the target distribution and the value is 1,0. Now, what is a cross entropy? Look at how we are doing cross entropy. There is some minus. Let us see what is

that minus later. We are taking one and we are combining with 0.875. One is from a P, 0.875 is from Q. And you are seeing $\log 2$, which means it is a surprise I of X.

[17:33] If it is I of X, you can either write it as minus or you can write it as 1 divided by. This concept is clear to us. We are taking one, which is from P of X. We are taking 0.875, which is from Q of X. Then you are taking both. So, and then substitute the values in this formula and then we have got a $H(P, Q)$ as 0.195. So, cross entropy is a denoted as $H(P, Q)$. And if you look into I've still not shown you the definition. I've still not shown you the formula. What we are doing here is we are taking the true distribution, we are taking the model's prediction also.

[18:20] Which means we are now combining two distributions. One is P, other one is Q. That is what is $H(P, Q)$. And how is the formula we have written? There is a minus. The minute you think of minus, it means it is I of X. Instead of writing it as 1 divided by, we are writing it as $\log$ minus $\log$ of. Now, you have $\log 2$ this. So, here this is nothing but Q of X. This term is Q of X. And this because of this it is I of X.

[18:54] Right? So, how your formula should look like? Is it not the formula should have something of P of X? Then it should have I of X. Then I of X you are going to combine with the Q of X. You'll have something like this. That we'll see in the next slide. Now, you have calculated and you have obtained to the value 0.193. What is this 0.193 we got? Is it not we got as a less value? Now, when you have a less, why did you get less value? You got less value because the actual outcome was sunny.

[19:28] And the model also has given highest probability to sunny only. Because the highest probability is given to the actual outcome, therefore, cross entropy value is less. Is it not this is the information we are inferring from this? Okay, let us look into another one. Still the outcome is sunny only, which means the original is 1 {comma} 0. Okay? And now model has a predicted this. What is that the model has predicted? This is for sunny, this is for rainy.

[20:01] Now, calculate the cross entropy. How is that you calculate the cross entropy? One, you get you will take this term. Zero, you will take this term. Because this entire term is zero, so we are considering only this term and we have got the value three. $H(P, Q)$, the value is three. » [clears throat] » When it is a three, what is the indication here? Is it not the value is very high? If the value is very high, what is the meaning? Is it not the model has assigned low probability to the actual outcome?

[20:39] The losses are large because the model has assigned a low probability to the correct class. Right? So, therefore, cross entropy what it is saying, cross entropy value is a small when the model assigns the high probability to the correct answer, and the cross entropy value is a large when it assigns low probability to the correct outcome. Right? Now, with this, let us see what is the formal definition or the explanation for a cross entropy? Cross entropy measures how well the predicted distribution

Q matches to the target distribution. Is it not if you go back?

[21:21] Is it not it is you are combine you are seeing model has a predicted something. Now you are combine you are seeing how well the model has a predicted. How is that you are checking? You are comparing it with the ground you know target distribution P. Therefore, cross entropy what it is doing, it is it measures how well the predicted distribution Q matches the target distribution P. Right? So, for a one outcome because now in our formula we have P of X sometime back we have seen. We have P of X, we have I of X also. So, if there is an I of X, it means it is for one outcome X what is that the model has a predicted.

[22:07] So, the model has a predicted and it has assigned the probability Q of X. So, then what will be its a surprise? Is it not the surprise is I of X and it is minus log two of Q of X. This is which you have already seen. Now, what exactly is a cross entropy? It checks how surprised the model is about the correct answer. Right? It checks how surprised the model is with the correct answer. So, this is what is the formula. We have P of X and it's a surprise I of X. If there is an I of X, can I write it as a minus? When you write it as a minus, can we now write minus Q of log Q of X?

[22:55] Are you now connecting it? In our KL divergence, we have a P of X log P of X divided by q of x. Is it not slowly we are approaching towards it? Can I also write instead of writing it as a minus, can I also write it as in this way? p of x log to 1 divided by q of x. We can even write in this way, right? So, what is the p of x? p of x is a true probability of outcome x. And what is i of x or minus log q of x? It is the surprise which is assigned to that outcome by the model. Therefore, we have these two terms. One is the true probability, second is the surprise that is assigned by the model. That is the model's prediction. Is the Combining all of this is the cross entropy.

[23:53] Right? So, not combining actually, it is measuring how well the model has a predicted when it is compared with the ground one p of x. Now, with this, let us see the formal derivation of KL divergence. So, in case for KL divergence, you require an cross entropy, you require an entropy, and in a both of this we have a self information. So, is it not from our previous slide we have a seen cross entropy and entropy? This is what we have obtained. What is a KL divergence?

[24:26] You are you know, you will take a cross entropy minus entropy. Now, substitute the values. Is it not h of pq? Is it not this value is written here? And minus h of p. h of p value is written. So, can we write minus of minus is plus here. So, is it not you get this a term? So, what did we do? This minus to minus we have converted to plus and then this minus you have as it is. If you look into both of here, what is that common we have? Is it not p of x is common? In the first term also, uh that is summation p of x, summation p of x.

[25:11] Is it not that these are the common here? So, um can I rearrange these terms? Why what is that you want to arrange? One is a negative plus you have positive. Can

I rearrange and then write the positive first, then a negative? Can we write in this way? What did we do? Nothing we have done. We after rearranging this term is written here and this has come here. So, I hope so far this is clear to you. What did we do? It is a cross entropy minus entropy.

[25:43] Just substitute the values. Minus of minus is a plus. So, first term is negative, second term is positive. Rearrange it. What is rearranging? Bring the second term to first term to second and second term to first. After rearranging we have got this. Now, can I take that common? So, here we have the common term. What is that common term? p of x . Can I take that out? Can when we take that p of x out, is it not it looks something like this? Uh summation p of x you have taken out, then you are left out with this term minus we are left out with this term. Is it not this? Okay, let me write in this way.

[26:26] Um right? We have got this. Now, what are we understanding? Is it not $\log a$ minus $\log b$? What is $\log a$ minus $\log b$? Is it not $\log a$ by b ? So, can I now substitute into this? Is it not we get this? p of x as it is. $\log a$ minus $\log b$ is $\log a$ by b . $\log p$ of x minus $\log q$ of x is $\log p$ of x by q of x . So, this is what is the overall KL divergence of formula. And what we have shown here is the forward KL. So, this is the KL divergence. This is one of the important concept. This is important because in our week three from next session onwards, we will be looking into the VAE and then the architecture. And we will be looking a detailed discussion on VAE encoder.

[27:24] So, in VAE loss function, we have something called as a KL divergence. And in week four, we'll be looking into advanced VAE concepts. There also we are using an KL divergence. So, therefore, the before we start the architecture of VAE and looking into an advanced VAE concepts, the whole intention of this session and then the previous session is to understand what is KL divergence and what is that formula is all about and how is that we have derived that formula. Thank you.